

An Overview of Force Measurement

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Abstract—The abstract goes here.

Index Terms—Interaction Force, Force Measurement.

I. INTRODUCTION

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August 26, 2015

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II. FORCE SENSING METHODES

There are many methods to measure forces exerted on objects. These methods differ from one another in the physical principle of sensing. In this section, we will explain some of these methods and compare between advantages and disadvantages of each method and the operating range of the mentioned transformers as shown in Table (I) and Table (II).

- A. Resistive force transducer
- B. Capacitive force transducer
- C. Inductive force transducer
- D. Piezoelectric force transducer
- E. Fiber optic force transducer
- F. Vibrating wire force transducer

This force transducer consists of a steel wire tensioned between two points. The center of the wire is subjected to a varying magnetic field. The magnetic field causes the wire to vibrate at a specific resonant frequency as shown in the Fig. (1a). When a force or strain is applied to the wire, its tension changes, resulting in a change in the vibration frequency. The frequency resulting from the wire's vibration is measured using a data logger or a digital readout [1]. The sensitivity of the vibrating-wire transducer, S_{vw} , which is defined as the ratio between the relative variation of frequency and the relative variation of wire length, which is the specific deformation ϵ [2]. This sensitivity reaches the value of 300 for VWFTs as compared to the 250 in case of the differential capacitor and with the 200 in case of the semiconductor (or only 2 for metallic) strain gauged transducers [3]. To achieve higher sensitivity, the differential wire gauge shown in the Fig. (1b) is used. This gauge consists of two pre-tensioned wires mounted within a rigid frame, with each wire excited by a pair of opposing magnets. When a force or strain is applied to the

central yoke, the tension increases in one wire and decreases in the other. Consequently, the frequency of one wire (f_1) rises while that of the other (f_2) falls. The difference between these two frequencies is directly proportional to the applied mechanical force [4]. These gauges are available in various types according [5] such as VWSG-A and VWSG-S for arc welding and spot welding to steel structures, respectively, and VWSG-E for embedment in concrete. Vibrating wire technology is widely used in geotechnical and structural engineering applications. In the literature, Simonetti [6] presented a research effort to adapt vibrating-wire force transducers for low-power applications and a new measurement technique based on the RMS value of the waveform response was discussed. Experimental results demonstrated that the proposed algorithm is particularly suitable for wireless sensor networks powered by energy harvesting solutions.

Six main methods were presented of detection force techniques. The operating principle of each method and the corresponding conditioning circuit have been explained. Additionally, the advantages and disadvantages of each method are outlined, along with the power range within which each of the aforementioned transducers can operate. In addition to the force-sensing technologies mentioned above, there are other methods for sensing force that are used in various specialized applications such as acoustic force transducers [17] [4] [2], gyroscopic force transducers [2] [4] [18] [19] [20]

III. STATIC VS DYNAMIC FORCE MEASUREMENT

IV. MULTI-DIMENSIONAL FORCE MEASUREMENT

V. FORCE ESTIMATION

VI. FORCE MEASUREMENT APPLICATIONS

- A. Force measurement in monitoring and testing tasks
- B. Use of force measurement in automated systems
- C. Force measurement in research and development laboratories

VII. CONCLUSION

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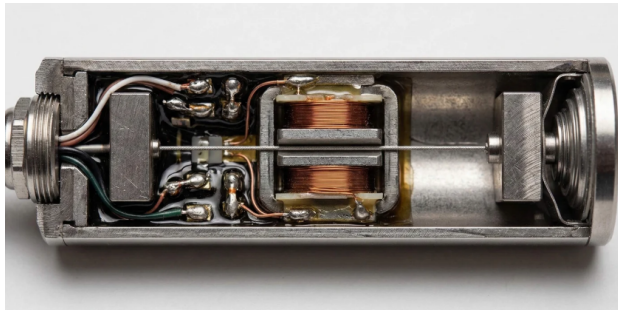
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TABLE I: Comparison of different measurement technologies [7]

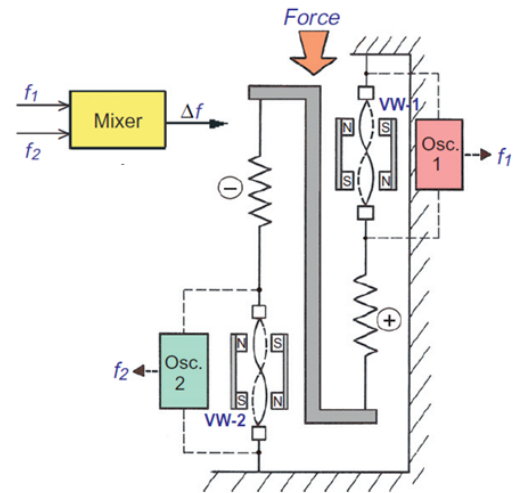
Detection technique	pros	Cons
Resistive	• simple construction and low cost • high and adjustable resolution • high reliability and maintenance-free • compatibility with VLSI • Thermal compensation using a Wheatstone bridge [8]. • Low hysteresis, good sensitivity [9] and a high signal-to-noise ratio at sub-newton force levels [10].	• higher power consumption. • damage caused by water entering the gauge [8] • The use of adhesive affects measurement accuracy [4, 10] • the gauge cannot be restored upon overloading [10] • rigid and fragile • scarce reproducibility • electromagnetic compatibility • narrow frequency bandwidth • contradictions between flexibility and sensitivity • damage arising from water ingress into the sensor [11]
Capacitive	• high sensitivity which allows for the measurement of small deviations [12] and resolution • large bandwidth • robustness • long-time stability • drift-free • durability • Adaptability to Environment • It is used to measure tactile information [13]. • Detection accuracy on the order of micronewtons or better [14].	• complex with the compulsory electronic circuits [15] • stray capacitance. • edge effect. • temperature sensitivity.
Inductive	• linear output • high power output • wide dynamic range	• lower frequency response • poor reliability • massive size.
Piezoelectric	• high frequency response • higher accuracy and finer resolution • high sensitivity and superior stiffness • high dynamic range [9, 4]. • They have instantaneous “zoom-in” capability (up to 105) [4].	• charge leakages • require to be embedded into the machining structure • deteriorations of voltages or drifts in the presence of static forces • poor spatial resolution • it is insensitive to the static force and it is easy subjected to the external electromagnetic noise [9].
FBG based system	• good reliability and high sensitivity • wide measurement range • noncontact and high-temperature performance • nonelectrical and immunity to electromagnetic interference and water damage • light weight and small size • intrinsically safe in the explosive environments • distributed measurement • ability to multiplex large numbers of FBGs along a single fiber [8].	• hard to construct dense arrays • more costly • complex Information processing system
vibrating wire	• Low power [16] • Zero drift, enhancing reliability and integrity [16]. • Its robust makes it less susceptible to mechanical damage, and it operates in humid environments [16]. • Its output is a frequency [4], allowing the signal to be transmitted over long cables without loss. • the vibrating wire gauge is compact and stable [4]. • High accuracy [4].	• complex devices [4]. • expensive mechanical devices [4].

TABLE II: Operating range of the force sensor according to its type [2]

force sensor type	Force measurement range
Resistive	$(10^{-13} \text{ to } 10^8) \text{ N}$
Capacitive	$(10^{-9} \text{ to } 10^4) \text{ N}$
Inductive	$(10^{-2} \text{ to } 10^6) \text{ N}$
Piezoelectric	$(10^{-1} \text{ to } 10^9) \text{ N}$
Vibrating wire	$(10^{-1} \text{ to } 10^7) \text{ N}$



(a)



(b)

Fig. 1: (a) Vibrating wire strain gauge [1], (b) differential vibrating wire strain gauge [4]

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